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REVIEW

Identification of Autism Spectrum Disorder-related Variants Through Integration of Transcriptome and Exome Geno Typing

Sayed AbdulAzeez

Department of Genetic Research, Institute for Research and Medical Consultations (IRMC), Imam Abdulrahman Bin Faisal University, Dammam, Saudi Arabia

★Corresponding Author : Sayed AbdulAzeez

Email : asayed@iau.edu.sa

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Abstract :

Autism is a complex disorder with genetic predisposition factors. Actual factors for treatment and early diagnosis have not yet been defined. In this study, we integrated transcriptome and exome genotyping to identify functional variants associated with autism spectrum disorders and their effects on gene expression, and to discover important mutations. More than 1800 patients were tested, and 70 (47 men, 23 women) with a mean age of 7.56 ± 3.68 years met DSM-5 criteria for autism. Analysis revealed that among putative functional exon variants examined (n = 243,345), 682 SNPs in 589 genes were significantly associated with autism ($p < 0.001$). The olfactory receptor gene on chromosome 6 was significant after Bonferroni correction ($\alpha = 0.05/243345 = 2.05 \times 10\text{-}7$), with a high degree of linkage disequilibrium at 6p22.1 ($p = 6.71 \times 10\text{-}9$). Genetic analysis of differential expression in autism patients compared to controls in total RNA-seq shows that is significantly upregulated (fold change ≥ 0.8 and p-value ≤ 0.05; n = 125) and downregulated (fold change ≦ -0.8 and p-value ≦ 0.05; n) was determined = 117) Gene. Regulatory variants associated with up-regulation (ITGB3BP) and down-regulation (DDR1 and MMP8) of genes in autism spectrum disorders through gene integration of significantly up- and down-regulated genes and significant SNPs (rs6657480, rs3130780, and rs1940475) were identified in people of Arab descent. The key mutations may serve as biomarkers to identify early autism in Arabs and help characterize genes involved in the mechanisms of susceptibility in autistic people.

Keywords : Autism spectrum disorder, Neurodevelopmental disorder Regulatory variants

Introduction :

Autism spectrum disorder (ASD) is a complex neurodevelopmental disorder with a wide range of symptoms of varying severity and has a high prevalence in the Gulf region. Current research suggests that genetic variations, changes in biological mechanisms, and the environment combine to influence the development of autism [1]. The most common symptoms of autism are poor social skills, communication difficulties, and repetitive movements. Autism can be detected as early as 3 years of age and in some cases as early as 18 months of age. The heritability of ASD is estimated to be between 40 and 90%, and in addition to epigenetic, environmental, and hormonal factors, many common mutations contribute to the large variation in the disease.

Additionally, the Diagnostic and Statistical Manual of Mental Disorders-Fifth Edition includes atypical sensory processing, which is reported as a diagnostic criterion in over 70% of patients with ASD [2].

Differentially expressed genes in autism are important for understanding associated signalling pathways. This meta-analysis examined 10 gene expression omnibus datasets containing 364 cases and 248 controls. 3105 genes were found to be consistently differentially expressed in autism (1680 down-regulated genes and 1425 up-regulated genes). PNPLA2, LYPLA2, LYPLA2P1, PLA2G4D, PLA2G6, PLA2G7, and PLA2G5 were also found to be associated with phospholipase A2 (PLA2) [3].

Gene Ontology enriched terms for molecular functions involved ribosome structural components and transcriptional regulatory activity, and Gene Ontology enriched terms for biological processes involved translation elongation and response to cytokine stimulation. The ribosomal pathway was the most important pathway in our KEGG study. Meta-analyses have discovered genes that are consistently differentially expressed in autism and biological pathways associated with changes in gene expression. Topological analysis of the autism brain identified a proteomic hub gene signature: CDK2, BAG3, MYC, TP53, HDAC1, CDKN1A, EZH2, GABARAPL1, TRAF1, and VIM. This study also revealed transcriptional regulators such as YY1, FOXL1, USF2, FOXC1, GATA2, NFIC, E2F1, NFKB1, TFAP2A, and HINFP [4].

Despite rapid developments in genetic technology, common variants of neuropsychiatric disorders have not yet been identified for diagnostic purposes. Exome microarrays provide a potential diagnostic method for many complex genetic diseases such as ASD. In contrast to other genetic methods, exome microarrays are suitable for profiling variants in associated genes with allele frequencies greater than 5%. Moreover, it provides exceptional coverage of the entire exonic region, provided by over 12,000 individual exome sequences. In this study, we performed total RNA sequencing and exome microarray analysis to identify the most common mutations associated with autism in Saudi patients [5].

Materials and methods:

Sample collection:

This study was conducted in accordance with the Declaration of Helsinki and was approved by the Institutional Review Board (IRB) of Imam Abdulrahman Bin Faisal University. Detection of cheek swab samples for the detection of DNA and RNA from children and adolescents up to 18 years of age in the Eastern Province of Saudi Arabia after obtaining signed informed consent from their parents and/or guardians to participate in the study A blood sample was collected. A total of 202 Arab subjects (132 controls and her 70 cases) provided cheek swab samples for genomic DNA extraction for exome genotyping (Table1). From January 2018 to December 2018, a total of 1,800 patients were screened and tested for autism at the pediatric neurology clinic. Of these, 70 of her patients (47 men, 23 women) with a mean age of 7.56 ± 3.68 years met her DSM-5 criteria for autism. Clinical evaluation of the patients revealed an overall positive family history of 8.57% (n = 6). Patients with comorbidities were excluded from this study. Healthy subjects without familial autism were selected as controls [6].

DNA extraction and exome array:

DNA was extracted from oral cell samples using the Gentra Puregene oral cell kit (Qiagen, Hilden, Germany). Illumina (San Diego, CA, USA) Human Exome Bead Chip Kit v1.0 and v1.1 consisting of 243,345 putative functional exon markers were used with his Illumina iScan for microarrays. All DNA samples were processed according to the manufacturer's protocols, and iScan control software (Illumina) was used for data collection. DNA extraction and microarray genotyping and analysis were performed at the Genetics Laboratory, Institute of Research and Healthcare Extension, Imam Abdulrahman Bin Faisal University, Dammam, Saudi Arabia. Genotyping was performed in this laboratory from 2016 to 2018 using the same chip and procedure [7].

Genotyping and useful analysis:

Genome Studio 2.0 data evaluation software program (Illumina) was used for the initial high-quality test of the call charge. a total of 21 controls and a pair of sufferers with autism had been excluded from the evaluation because of a name charge <0.975%, and the consequences had been hence re-clustered. The Hardy- Weinberg equilibrium (HWE) changed into examined one at a time most of the case and manage corporations with a 1 diploma of freedom (df) genotypic chi-square take a look at.

Differences in clinical characteristics between cases and controls have been calculated by using the 2-pattern t-check or the χ2 check, as appropriate, the usage of IBM SPSS information model 23 software. Kaviar and SNP-Nexus were used to verify variations pronounced at a base pair role at the respective chromosome as in step with Genome Reference Consortium Human construct 37 (GRCh37.p13.) [8].

Case-control association analyses, odds ratios, and 95% confidence periods have been calculated to assess the consequences of different alleles and haplotypes the use of Haploview model 4.2 and gPLINK version 2.050. Bonferroni corrections or false dis over fee corrections were applied to accurate the p-values of the 243,345 single nucleotide polymorphisms (SNPs) (corrected $\alpha = 0.05/243,345 = 2.05 \times 10\text{-}7$) to control inflation of the kind I blunders price.

A p-price < 0.05 changed into considered full-size. The surprisingly large ($p < 1 \times 10\text{-}5$) genes have been annotated for useful implications the use of DAVID 6.7, STRING 10.5, SNPnexus Expression Atlas Reactome, FunRich, Toppgene and Enrichr [8].

Complete RNA Sequencing and Differentially Expressed Genes:

Overall ribonucleic acid (RNA) become extracted from blood (30 controls and 30 sufferers with autism) the use of the RNeasy plus micro kit (Qiagen, Hilden, Germany) following the manufacturer's guidelines. General RNA become quantitated the usage of the Qubit RNA HS assay kit (existence technology, Eugene, USA), and the first-class turned into assessed by using an Agilent 2100 (Agilent, Santa Clara, CA, america). High-quality RNA changed into used to assemble the cDNA library [1].

Twenty milligrams of total RNA from each sample became handled one at a time with oligo (dT) for mRNA enrichment. N6 random primers had been used for opposite transcribing the goal RNA fragments into double-strand cDNA. Stop restore turned into done on double-strand cDNA fragments, and adaptors have been ligated to the double-strand cDNA [9].

All the ligated double-strand cDNA had been PCR-amplified, denatured by using warmth into unmarried strand DNA and quantified, and subjected to DNA nanoball synthesis and sequencing on the DNBseq platform [5].

Transcriptome meeting become finished after subjecting the raw reads to nice manage. Reads of adaptors and unknown bases, as well as low-excellent reads, were discarded. The raw RNA sequencing information reads with excessive fine have been mapped the usage of the tophat RNA series mapping tool. The aligned sequence facts have been analyzed for the differential expression between the autistic and the control subjects by using thinking about the foldchange ≥ 0.8 and p-price ≤ 0.05 for upregulated genes and the foldchange ≤ -0.8 and p-cost ≤ 0.05 for downregulated genes.

All of the expressed genes from the autistic and manage samples with a first-rate Phred score ≥ 30 were handiest subjected to the differential expressed gene evaluation. The exceptionally considerably up- and downregulated genes were annotated for pathway enrichment analysis using DAVID 6.7, STRING 10.5, Expression Atlas, Reactome, FunRich, Toppgene, and Enrichr [10].

Results:

The observe discovered the considerable regulatory editions related to the increasement of autism in humans with Arab ancestries the usage of multiomic evaluation (figure 1).

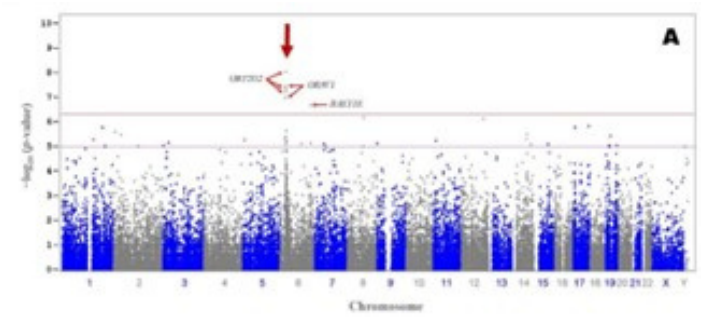

Figure 1: The log10 (p) values from the association are plotted.

A total of 13 SNPs have been substantially (p < 2.05 × 10-7) related to ASD even after Bonferroni or false discovery price corrections (corrected α = 0.05/243,345 = 2.05 × 10-7) a number of the 243,345 SNPs screened, and that they obeyed the HWE table 2 summarizes the maximum large SNPs that had been related to autism in patients of Arab ancestry not unusual variations in olfactory receptor circle of relatives 12 subfamily D member 2 (OR12D2) and olfactory receptor circle of relatives five subfamily V member 1 (OR5V1) showed the most powerful signal in Arab sufferers with autism [7].

The pinnacle 10 most significantly (p < 2.05 × 10-7) associated SNPs had been within the vicinity of the olfactory receptor genes OR12D2 and OR5V1 on chromosome 6p22.1. Extra statistics approximately the huge SNPs (p < 0.0001) in autism patients is furnished in table S1. The effects advise that a unmarried gene, OR12D2 (inclusive of up and downstream; NC_000006.11:g.29360183-g.29369519), has 10 commonplace SNPs with large effects on risk (rs9257819, rs2022077, rs9257834, rs4987411, rs2073154, rs2073153, rs2073151, rs2073149, rs1028411, and rs2394607), together with 5 SNPs within the coding place (rs9257834, rs4987411, rs2073154, rs2073153, and rs2073151). Consistent with variation function on every chromosome [11].

Positions of candidate genes of practical exonic versions at olfactory receptor genes on chromosome 6p22.1 for autism are indicated by way of red arrows. The horizontal red line indicates the preset threshold of p=4.00 × 10-7. The horizontal purple line indicates the suggestive threshold of p=1.00 × 10-5. B: Genetic affiliation of the maximum enormous variants on the OR12D2 and OR5V1 genes in the chromosome 6p22.1 location and linkage disequilibrium results in subjects of Arab ancestry with autism. Block 1 indicates the most significant danger haplotype, AAGTCTGATT (p = 6.7082 × 10-9), associated with the Arab-ancestry autism subjects [12].

All the association exams were screened using the frequency of teen alleles in controls, the p-value of the HWE, and the kind I error price to identify the strongest genetic findings, which had been enter for linkage disequilibrium in HapMap SNPs of selected areas on chromosome 6 (figure 2) [13].

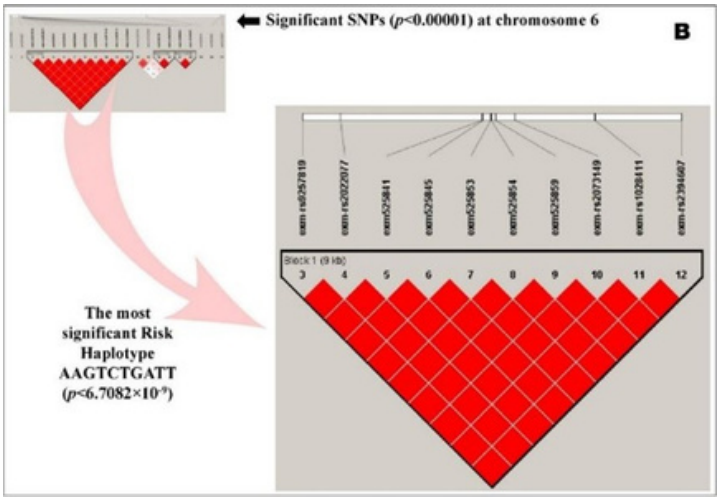

Figure 2: A: Manhattan plot of putative functional exonic variants n = 243,345.

The variants within the place of the olfactory receptor genes OR12D2 and OR5V1 had a high degree of linkage disequilibrium (p = 6.71 × 10-9). The haplotype related to the hugest risk in sufferers of Arab ancestry with autism, AAGTCTGATT (p = 6.7082 × 10–nine), consisted of the top 10 SNPs (rs9257819A, rs2022077A, rs9257834G, rs4987411T, rs2073154C, rs2073153T, rs2073151G, rs2073149A, rs1028411T, and rs2394607T) [3]. the protecting haplotype maximum significantly related to the manage subjects of Arab ancestry, CTTCGGATGC (p = 2.43 × 10-8) also consisted of the top 10 SNPs (rs9257819C, rs2022077T, rs9257834T, rs4987411C, rs2073154G, rs2073153G, rs2073151A, rs2073149T, rs1028411G, and rs2394607C) with the alleles that had been now not related to autism (table 1) [14].

Block	Haplotype	Frequency	Case, Control Ratio Counts	Case, Control Frequencies	Chi-Square	*p*-value
Block 1	AAGTCTGATT	0.547	97.0:33.0, 89.0:121.0	0.746, 0.424	33.618	6.71×10^{-9}**
	CTTCGGATGC	0.421	30.0:100.0, 113.0:97.0	0.231, 0.538	31.12	2.43×10^{-8}*
	AAGTCTGTTC	0.024	3.0:127.0, 5.0:205.0	0.023, 0.024	0.002	0.9655
Block 2	CG	0.653	102.0:28.0, 120.0:90.0	0.785, 0.571	16.104	6.00×10^{-5}**
	AT	0.347	28.0:102.0, 90.0:120.0	0.215, 0.429	16.104	6.00×10^{-5}*
Block 3	TA	0.541	88.0:42.0, 96.0:114.0	0.677, 0.457	15.62	7.74×10^{-5}**
	CG	0.459	42.0:88.0, 114.0:96.0	0.323, 0.543	15.62	7.74×10^{-5}*

** Significant risk haplotypes associated with autistic patients. * Significant protective haplotypes associated with control subjects.

Table 1: Significant haplotypes of SNPs on chromosome 6 in autistic patients.

Blood samples had been accumulated inside the RNA protectant vacutainer with an RNA stabilizer to acquire greater correct consequences. General RNA turned into isolated for the transcriptome profiling look at using paired-quit sequencing. Raw reads had been checked for a quality Phred score ≥ 30. The qualifying reads were processed for mapping on the reference genome and expression profile preparation. The uncooked RNA sequencing data reads have been trimmed for the adaptors and mapped by means of the usage of the tophat RNA series mapping device with the reference RNA collection [10]. All RNA series data were subjected to finish differential expression analysis among the autistic and the manage subjects [15]. The differential expression between the autistic and manipulate topics turned into calculated, revealing upregulated (foldchange ≥ 0.8 and p-value ≤ 0.05) and downregulated (foldchange ≤ -0.8 and p-fee ≤ 0.05) genes. Autistic sufferers and wholesome controls have been taken into consideration two wonderful corporations for differential expression calculation the usage of Cuffdiff. Genes were filtered for differential expression thru all good sized DEGs primarily based on a p-value ≤ 0.05. For upregulated genes, we used a cutoff p-fee ≤ 0.05 and foldchange ≥ 0.8, and for downregulated genes, we used a cutoff p-price ≤ 0.05 and foldchange ≤ -0.8. All of the expressed genes from the autistic and control samples with a fine Phred rating ≥ 30 had been subjected to differential expressed gene analysis, and the consequences are provided in discern 3 as volcano plots. The extensively upregulated and downregulated genes were taken into consideration with foldchange ≥ 0.8 and p-cost ≤ 0.05 and foldchange ≤-0.8 and p-price ≤ 0.05, respectively [16].

All the notably upregulated and downregulated genes had been looked for all through the go and pathway enrichment evaluation. Notably upregulated genes and their interactions revealed quite sizable protein–protein interactions based on the predictive version enrichment (p-cost: < 1.0 × 10-16). Downregulated genes (p-value <0.001) and their interactions verified sizable PPI enrichment with a p-fee = 2.05 × 10-6 [5]. Pathway enrichment through the substantially upregulated and downregulated genes. Ribosome, oxidative phosphorylation, protein export, Parkinson's disorder, and Alzheimer's ailment-related pathways were considerably enriched by the upregulated genes. The impact on the genetic information processing and translation turned into substantially enriched by the 17 upregulated genes via the ribosome pathway.

The HIF-1 signaling pathway, pyruvate metabolism, propanoate metabolism, metabolic pathways, the rap1 signaling pathway, the AMPK signaling routeway, the glucagon signaling pathway, proteoglycans in most cancers, renal cellular carcinoma, fatty acid biosynthesis, the insulin signaling pathway, and EGFR tyrosine kinase inhibitor resistance had been substantially enriched via the downregulated genes (table 6).

Integration of Transcriptome and Exome Genotyping:

Genotyping and exome array analysis the various 243,345 putative useful exonic variations revealed 589 genes considerably ($p < 0.001$) associated with autism, with 682 significant variants. Furthermore, the differentially expressed gene analysis based on transcriptome sequencing for the autistic patients compared to controls identified significantly upregulated (foldchange ≥ 0.8 and p-value ≤ 0.05; n = 125) and downregulated (foldchange ≤-0.8 and p-value ≤ 0.05; n = 117) genes [17]. The integration of significantly up- and downregulated genes and genes of significant SNPs using Venn diagram analysis identified three regulatory variants (rs6657480, rs3130780, and rs1940475) associated with upregulated (rs6657480, p = 0.000762, (NC_000001.10: g.63999868T > C) nearest upstream to the gene ITGB3BP) and downregulated (rs3130780, p = three.12 × 10-5 (NC_000006.11: g.30874308T > G) nearest upstream gene, DDR1; and rs1940475, p = 0.0001807 (NC_000011.9:g.102593248T > C; NM_002424.3:c.259A > G; NP_002415.1:p.Lys87Glu, MMP8)) genes in autism spectrum disease in sufferers of Arab ancestries [18,19].

Discussion:

This observe completed a complete research of the genetic structure of ASD using an exome microarray evaluation and RNA sequencing. We diagnosed a hundred and 55 significantly ($p < 0.0001$) related SNPs, and the most strongly ($p < 2.05 \times 10\text{-}7$) related 13 SNPs were investigated further not one of the thirteen SNPs had formerly been stated in Saudis with autism. The OR12D2 olfactory receptor gene, which codes for the G-seasonedtein coupled receptor (GPCR), includes the sign to sensory neurons to trigger scent and has 10 commonplace variations and huge effects on threat, turned into associated with autism in Arabancestry topics and showed an excessive diploma of linkage disequilibrium ($p = 6.71 \times 10\text{-}9$) on the enormous threat haplotype AAGTCTGATT.

The OR12D2 gene (gene expression for OR12D2: ENSG00000168787.4) was overexpressed inside the mind (hypothalamus ×13.9, caudate ×10.7, hippocampus × 8.2 nucleus accumbens × 5.0 and cervix-endocervix × 4.6).

Some studies have proposed uncommon sensory processing, mainly olfactory processing, in some of neurodevelopmental issues, along with autism. Confirmed that the OR2L13 G-protein, joined by way of the olfactory receptor, that is involved in resetting the neuronal reaction to smells, is especially ordinary in autism in phrases of DNA methylation and expression, suggesting a likely motive for olfactory dysfunction in ASD. The olfactory receptor genes OR12D2 and OR5V1, which might be involved within the olfactory signaling pathway, G alpha (s) signaling occasions, GPCR downstream signaling, and GPCR signaling and signal transduction together modify the overall olfactory increase, behavior, and impairment associated with autism. A purposeful evaluation of the genes [4]. The sizable genes on this take a look at blanketed integrin subunit alpha eleven (ITGA11), that's related to interest deficit.

Hyperactivity disorder and interest deficit disease with hyperactivity; MHC elegance I polypeptide-related series A (MICA), related to signs of inflammatory bowel ailment, chronic ulcerative colitis, and Crohn's ailment; bromodomain-containing 2 (BRD2), which has been related to Alzheimer's disease and myoclonic epilepsy; mobile division cycle 14C pseudogene (CDC14C), which is related to Parkinson's sickness and sleep; methylenetetrahydrofolate dehydrogenase (NADP + structured) 1-like (MTHFD1L), which is associated with stroke and Alzheimer's sickness; myosin IXB (MYO9B), related to symptoms of celiac sickness, inflammatory bowel disease, persistent ulcerative colitis a couple of sclerosis, and schizophrenia; and ring finger protein 144A (RNF144A), which has been related to depressive disorder, schizophrenia, and stroke [9]. As an instance, SNP rs2727943 is extensively correlated with ASD and has been suggested in bipolar cases as an intergenic version that occurs between the large-2 and CNTN6 genes both of which produce a neuronal membrane protein that paperwork the axonal connections within the growing apprehensive machine. Unlike previous results, our study identified a single gene, OR12D2 (consisting of up and downstream; NC_000006.11:g.29360183-g.29369519), with 10 common SNPs and big consequences on danger in patients with autism most preceding consequences have cautioned that no single gene has such massive consequences on chance ($p = 6.71 \times 10\text{-}9$).

A comprehensive review on the genetics of ASD defined the function of various genes and their medical relevance to autism [5, 10]. Right here, we describe an extra gene, OR12D2, that ought to be considered for molecular pathway research in the destiny. The enrichment evaluation of the genes nearest upstream/nearest downstream genes of the surprisingly associated SNPs using Enrichr in GWAS Catalogue 2019 revealed enormous sicknesses drug terms associated with ASD or schizophrenia (p = 0.001107; adjusted p = 0.0321) on 8 genes, namely SFTA2, BRD2, HCP5, OR5V1, HLA-DMA, ZKSCAN8, OR12D2 and MICA. Genes (TMEM185B MTHFD1L, QPCT and H2AFV, p = 0.0098) have been downregulated relative to their median expression within the ventral juxtacommissural pretectal nucleus some of the genes nearest upcirculate nearest downstream genes of highly related SNPs the usage of Enrichr. Kids with systemic lupus erythematosus (SLE) have a high danger of developing autism within the gift have a look at, seven genes (BRD2, OR12D2, CR2, OR5V1, SFTA2, HCP5, and HLA-DMA) have been rather huge for SLE (p = 0.000029; Bonferroni = 0.0041) the usage of DAVID.

The chance haplotype of the OR12D2 gene in autism subjects and the overexpression of OR12D2 inside the brain, dysfunction in neurodegenerative tactics, olfactory impairment, and the overuse of aroma compounds or vital oils, flowers, aromatic wooden, herbs, frankincense, and perfumes a few of the Arab populace provide clues that olfactory sensory deficits and overexpression of OR12D2 must be studied in element many of the Arab populace to expose the underlying etiology of autism.

Conclusion:

The Saudi autistic sufferers are considerably distinctive according to the gene expression profile. The fivefold downregulated MPO, TMCC3, MMP8, CA1, and ELF3 genes and the fivefold upregulated MTRNR2L8 gene can be considered for the early identity of autism patients among Saudis. This look at recognized the OR12D2 gene, such as the 5' and 3' regions related to 10 ASD SNPs on chromosome 6p22.1.

The SNPs detected in patients with autism within the modern examine are variations in genes that both control fundamental mobile survival strategies or direct neurological capabilities that might make a contribution to the severity of ASD. Screening the molecular defects which might be well-known in Saudi autistic instances is a completely important step to better understand the viable genes which are concerned in the improvement of ASD. Regulatory versions (rs6657480, rs3130780, and rs1940475) diagnosed the up- (ITGB3BP) and downregulation (DDR1 and MMP8) of genes in autism spectrum ailment in patients of Arab ancestry found through the combination of up and downregulated genes and genes of tremendous genotypes. Similarly, studies are needed to analyse the pathogenic pathways of the mentioned SNPs and genes to broaden potential therapeutic objectives within the destiny. The presence of a panel of the maximum popular variations in ASD instances will useful resource in improving our know-how and expertise of the complexity in the back of the variety in ASD.

Conflict of interest:

The authors declare no conflict of interest.

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